JavaScript is disabled on your browser.

compact1, compact2, compact3

java.util.jar

Class Pack200

java.lang.Object
    java.util.jar.Pack200

```
public abstract class Pack200
extends Object
```

Transforms a JAR file to or from a packed stream in Pack200 format. Please refer to Network Transfer Format JSR 200 Specification at http://jcp.org/aboutJava/communityprocess/review/jsr200/index.html

Typically the packer engine is used by application developers to deploy or host JAR files on a website. The unpacker engine is used by deployment applications to transform the byte-stream back to JAR format.

Here is an example using packer and unpacker:

```
import java.util.jar.Pack200;
import java.util.jar.Pack200.*;
...
// Create the Packer object
Packer packer = Pack200.newPacker();

// Initialize the state by setting the desired properties
Map p = packer.properties();
// take more time choosing codings for better compression
p.put(Packer.EFFORT, "7");  // default is "5"
// use largest-possible archive segments (>10% better compression).
p.put(Packer.SEGMENT_LIMIT, "-1");
// reorder files for better compression.
p.put(Packer.KEEP_FILE_ORDER, Packer.FALSE);
// smear modification times to a single value.
```

```
p.put(Packer.MODIFICATION_TIME, Packer.LATEST);
// ignore all JAR deflation requests,
// transmitting a single request to use "store" mode.
p.put(Packer.DEFLATE_HINT, Packer.FALSE);
// discard debug attributes
p.put(Packer.CODE_ATTRIBUTE_PFX+"LineNumberTable", Packer.STRIP);
// throw an error if an attribute is unrecognized
p.put(Packer.UNKNOWN_ATTRIBUTE, Packer.ERROR);
// pass one class file uncompressed:
p.put(Packer.PASS_FILE_PFX+0, "mutants/Rogue.class");
try {
    JarFile jarFile = new JarFile("/tmp/testref.jar");
    FileOutputStream fos = new FileOutputStream("/tmp/test.pack");
    // Call the packer
    packer.pack(jarFile, fos);
    jarFile.close();
    fos.close();

    File f = new File("/tmp/test.pack");
    FileOutputStream fostream = new FileOutputStream("/tmp/test.jar");
    JarOutputStream jostream = new JarOutputStream(fostream);
    Unpacker unpacker = Pack200.newUnpacker();
    // Call the unpacker
    unpacker.unpack(f, jostream);
    // Must explicitly close the output.
    jostream.close();
} catch (IOException ioe) {
    ioe.printStackTrace();
}
```

A Pack200 file compressed with gzip can be hosted on HTTP/1.1 web servers. The deployment applications can use "Accept-Encoding=pack200-gzip". This indicates to the server that the client application desires a version of the file encoded with Pack200 and further compressed with gzip. Please refer to Java Deployment Guide for more details and techniques.

Unless otherwise noted, passing a `null` argument to a constructor or method in this class will cause a `NullPointerException` to be thrown.

Since:

1.5

Nested Class Summary

Nested Classes

Modifier and Type	Class	Description

static interface	Pack200.Packer	The packer engine applies various transformations to the input JAR file, making the pack stream highly compressible by a compressor such as gzip or zip.
static interface	Pack200.Unpacker	The unpacker engine converts the packed stream to a JAR file.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
static Pack200.Packer	newPacker()	Obtain new instance of a class that implements Packer.
static Pack200.Unpacker	newUnpacker()	Obtain new instance of a class that implements Unpacker.

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Method Detail

newPacker

```
public static Pack200.Packer newPacker()
```

Obtain new instance of a class that implements Packer.

- If the system property `java.util.jar.Pack200.Packer` is defined, then the value is taken to be the fully-qualified name of a concrete implementation class, which must implement Packer. This class is loaded and instantiated. If this process fails then an unspecified error is thrown.
- If an implementation has not been specified with the system property, then the system-default implementation class is instantiated, and the result is returned.

Note: The returned object is not guaranteed to operate correctly if multiple threads use it at the same time. A multi-threaded application should either allocate multiple packer engines, or else serialize use of one engine with a lock.

Returns:

`A newly allocated Packer engine.`

newUnpacker

```
public static Pack200.Unpacker newUnpacker()
```

Obtain new instance of a class that implements Unpacker.

- If the system property `java.util.jar.Pack200.Unpacker` is defined, then the value is taken to be the fully-qualified name of a concrete implementation class, which must implement Unpacker. The class is loaded and instantiated. If this process fails then an unspecified error is thrown.
- If an implementation has not been specified with the system property, then the system-default implementation class is instantiated, and the result is returned.

Note: The returned object is not guaranteed to operate correctly if multiple threads use it at the same time. A multi-threaded application should either allocate multiple unpacker engines, or else serialize use of one engine with a lock.

Returns:

`A newly allocated Unpacker engine.`

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